

Lecture Notes 2

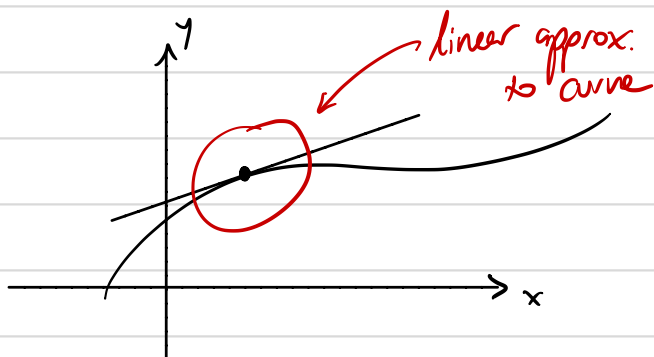
Linearity

"Calculus is the art of approximating some curvy things w/ some straight things." - William Burke.

Linear models are unrealistic (!) so why study them?!

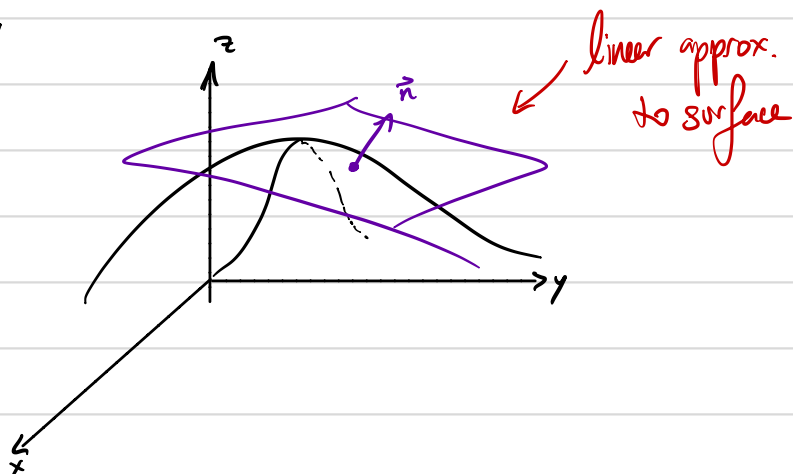
- Linear models are easier.
- Linear models are fine approximations to realistic and nonlinear models.
- Linear models allow us to solve problems, and therefore progress.

Ex //



$$\underline{ax + by = c}$$

Ex //



$$ax + by + cz = d$$

Ex // $m\ddot{\theta} = -mg \sin \theta \Rightarrow \ddot{\theta} = -\frac{g}{l} \sin \theta$

↳ $\ddot{\theta} = -\frac{g}{l} \theta$ ↙ solvable for small θ

Ex// $0.6 \bar{Q} + 0.4 \bar{F} = \text{Score}$ is a linear equation!

Defn: An equation $F(x_1, x_2, \dots, x_n) = C$ is linear if it is of the form

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = C,$$

where $a_1, a_2, \dots, a_n$, and C are constants.

If $C=0$, the linear equation is called homogeneous.

Homogeneous linear equations play a very special role in math.

Ex// $ax + by = 0$ is homogeneous. Graph an example, what do you notice?

Ex// $ax + by + cz = 0$ is homogeneous. Graph an example, what do you notice?

In the context of ODEs, constants are taken up by functions of t .

Each x_i is some $y^{(i)}$, a function also of t .

Defn: Linear Differential Equation

An ODE $F(y, y', \dots, y^{(n)}) = f(t)$ is linear if it is of the form:

$$a_n(t) \frac{d^n y}{dt^n} + a_{n-1}(t) \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_1(t) \frac{dy}{dt} + a_0(t) y = f(t)$$

where all $a_i(t)$ and $f(t)$ are defined over some common interval.

If $f(t)=0$ over the aforementioned interval, the ODE is called homogeneous.

Ex// All first-order linear ODEs can be written as $y' + p(t)y = f(t)$

Ex// $(y')^2 + p(t)y = f(t)$ is non-linear since y & y' do not appear w/ only exponent 1.

Ex// $y' + p(t)y^2 = f(t)$ is also non-linear for the same reason.

Ex// $y'' + p(t)y' + q(t)y = f(t)$ is a linear second-order non-homogeneous ODE.

Linearity & non-linearity have nothing to do w/ $f(t)$, but everything to do w/ the dependent variables $y, y', \dots, y^{(n)}$.

Linearity: $y, y', y'', \dots, y^{(n)}$ appear together, unmultiplied by each other, and w/ exponent 1.

Non-linearity: Not linear ↗

Ex// $y \cdot y' + y^2 = \sin(t)$.

$\sin(t)$ irrelevant for linearity.

Since $y \cdot y'$, this is non-linear!

"Abstraction"

Theory of Linear Operators

Consider the equations: $a_1 x_1 + a_2 x_2 + \dots + a_n x_n = C$

$$a_n(t) y^{(n)} + a_{n-1}(t) y^{(n-1)} + \dots + a_1(t) y' + a_0(t) y = f(t)$$

The coefficients may change from situation to situation, but the collection of objects

$$[x_1, x_2, \dots, x_n] \neq [y^{(n)}, y^{(n-1)}, \dots, y', y]$$

does not change its structure.

We call these collections of objects vectors. Using this notation we can recast all of the above into a unified language. Label vectors as $\vec{x}$ or $\vec{y}$ as the situation warrants.

Definition: A numerical function $L(\vec{x})$ of a vector argument is called a linear operator if it satisfies the following two conditions:

- ① $L(k\vec{y}) = k L(\vec{y})$ for all $k \in \mathbb{R}$
- ② $L(\vec{x} + \vec{y}) = L(\vec{x}) + L(\vec{y})$

Ex// The derivative operator is a linear operator:

$$\frac{d}{dx}(k f(x)) = k \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x))$$

Ex// The integral operator is a linear operator:

$$\int k f(x) dx = k \int f(x) dx$$

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx$$

From the PoV of Linear Algebra, they are the same sort of object!

Ex// More relevant: Given $y' + \sin(t)y = \cos(t)$

$$L(y) = \left(\frac{d}{dt} + \sin(t)\right)y = \cos(t)$$

The linear operator is $\frac{d}{dt} + \sin(t)$. It's not multiplying y , it's acting on y !

Ex// $t^2 y'' + t y' + y = 0$; $L = t^2 \frac{d^2}{dt^2} + t \frac{d}{dt} + 1$

so that the ODE can be written $L(y) = 0$.

Ex// If $\vec{x} = [x_1, x_2, \dots, x_n]$, then

$\langle \vec{a}, \vec{x} \rangle = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$ is a linear operator called the inner product!

can prove easily (do it in prep. for what is coming)

① $\langle \vec{a}, k\vec{x} \rangle = k \langle \vec{a}, \vec{x} \rangle$

② $\langle \vec{a}, \vec{x} + \vec{y} \rangle = \langle \vec{a}, \vec{x} \rangle + \langle \vec{a}, \vec{y} \rangle$

This all leads to a fundamentally important idea w/ wide applications!

The Superposition Principle for Linear Homogeneous Equations

let $\vec{u}_1$ & $\vec{u}_2$ be solutions of the linear homogeneous eqn
 $L(\vec{u}) = 0$.

Then their sum $\vec{u} = \vec{u}_1 + \vec{u}_2$ is also a solution; and
a multiple $\vec{u} = k\vec{u}_1$ is also a solution.

(proof follows from the definition of a linear operator)

Ex/ Given $y'' + 4y = 0$, two solutions are $y_1 = \sin(2t)$ & $y_2 = \cos(2t)$

$$\begin{array}{ll} \text{pf: } y_1 = \sin(2t) & y_2 = \cos(2t) \\ y_1' = 2\cos(2t) & y_2' = -2\sin(2t) \\ y_1'' = -4\sin(2t) & y_2'' = -4\cos(2t) \end{array}$$

$$y_1'' + 4y_1 = -4\sin(2t) + 4\sin(2t) = 0 \checkmark$$

Both y_1 & y_2
solve the ODE!

$$y_2'' + 4y_2 = -4\cos(2t) + 4\cos(2t) = 0 \checkmark$$

Superposition dictates that $y = y_1 + y_2$ also solves the ODE!
and that ky_1 (or ky_2) also solves the ODE!

$$\begin{aligned} y'' + 4y &= (y_1 + y_2)'' + 4(y_1 + y_2) = \underbrace{y_1'' + y_2''}_{\text{why?!}} + 4y_1 + 4y_2 = [y_1'' + 4y_1] + [y_2'' + 4y_2] \\ &= 0 + 0 = 0 \checkmark \end{aligned}$$

$$\begin{aligned} \text{likewise, } y'' + 4y &= (ky_1)'' + 4(ky_1) = \underbrace{k y_1''}_{\text{why?!}} + k(4y_1) = k[y_1'' + 4y_1] = k \cdot 0 \\ &= 0 \checkmark \end{aligned}$$

Nonhomogeneous Principle

Let $\vec{u}$ be any solution (the particular solution) to a linear nonhomogeneous equation

$$L(\vec{u}) = C \quad (\text{if algebraic})$$

or

$$L(\vec{u}) = f(t) \quad (\text{if differential})$$

Let $\vec{u}_h$ be a solution to the associated linear homogeneous equation

$$L(\vec{u}_h) = 0.$$

Then $\vec{u} = \vec{u}_h + \vec{u}_p$ is also a solution to the nonhomogeneous equation

$$L(\vec{u}) = C \quad \text{or} \quad L(\vec{u}) = f(t).$$

Furthermore, every solution to the nonhomogeneous equation must be of the form

$$\vec{u} = \vec{u}_h + \vec{u}_p$$

ef:

This is simply a retelling of definitions:

$$L(\vec{u}) = L(\vec{u}_h + \vec{u}_p) = L(\vec{u}_h) + L(\vec{u}_p) = 0 + \left\{ \begin{matrix} C \\ f(t) \end{matrix} \right\}$$

depending on the equation.

The proof for the second part is left to you. (Book does it)

Very important observation: All solutions to $L(\vec{u}) = f(t)$ (or C)

must be of the form $\vec{u} = \vec{u}_h + \vec{u}_p$

↑
the part that
gives zero

↑
the part that
gives $f(t)$ (or C)

This concept appears
again & again! In
fact, you've seen
it before!

$$\text{Ex// } \frac{d}{dt}(y) = \cos(t)$$

$$\text{Then } y = y_h + y_p$$

To find y_p , we observe that $\frac{d}{dt}(\sin(t)) = \cos(t)$, so we conclude $y_p = \sin(t)$!

To find y_h , we observe that $\frac{d}{dt}(y_h) = 0$ means $y_h = \text{const} = C$

$$\text{so the solution to } \frac{d}{dt}(y) = \cos(t) \text{ is } y = \underset{\substack{\uparrow \\ y_p}}{\sin(t)} + \underset{\substack{\uparrow \\ y_h}}{C} \quad (!)$$

Solving all first-order linear non-homogeneous ODEs!

To start off, assume an equation of the form $y' + p(t)y = q(t)$. All first-order linear ODEs can be written that way, simply divide any coeff. of y' on both sides of the ODE and viola!

There are two methods presented in the book. The first method is too much clutter, unhelpful, and far too confusing! We will only discuss the easier, more efficient, and sensible method called:

The Integrating Factor Method

The integrating factor method depends entirely on one simple observation: that the LHS of the first-order linear (non)homogeneous ODE is the result of _____
Fill in the blank.

Observe: $y' + p(t)y = q(t)$

It's almost something recognizable! There's only one bit missing.

ex// $ty' + y = \text{something}$... LHS is the derivative of the function ty wrt t !

Can we always have that form? YES!

If we multiply everything by something called an integrating factor:

$$\mu = e^{\int p(t) dt} \quad \text{then observe that}$$

$$\mu' = p(t) e^{\int p(t) dt} = p(t) \mu(t)$$

→

We then get

$$\mu(t) y' + p(t) \mu(t) y = q(t) \mu(t)$$

The LHS is again the derivative of $\mu(t) y$ (!)

$$\begin{aligned} \therefore \frac{d}{dt} (\mu(t) y) &= \mu(t) y' + \mu'(t) y \\ &= \mu(t) y' + p(t) \mu(t) y \quad (!) \end{aligned}$$

How to solve the actual ODE

① Find your integrating factor $\mu(t) = e^{\int p(t) dt}$

② Multiply your ODE (both sides) by $\mu(t)$.

③ LHS will always be $\frac{d}{dt} (\mu y)$, so integrate both sides to isolate y .

④ Isolate y .

Ex// §2.2 #6 $\frac{dy}{dt} + 2ty = t$

The ODE is in the correct form. $p(t) = 2t$, $q(t) = t$. Find the integrating factor and get to work!

$$\mu(t) = e^{\int p(t) dt} = e^{\int 2t dt} = e^{t^2}$$

we left out the "+C" ... why?

Multiply everything by μ :

$$e^{t^2} y' + 2t e^{t^2} y = t e^{t^2}$$

ODE can be rewritten: $\frac{d}{dt}(e^{t^2} y) = t e^{t^2}$

Now integrate!

$$e^{t^2} y = \int t e^{t^2} dt + C$$

$$e^{t^2} y = \frac{1}{2} e^{t^2} + C, \text{ and now isolate } y!$$

$$y = \frac{1}{2} + C e^{-\frac{1}{2} t^2} \dots \text{and that's it!}$$

Hold on! The Nonhomogeneous principle said all linear ODE can be written as $y = y_h + y_p$, are we really done?!

YES! It turns out the part w/o the const. of integration is your particular solution, and the homogeneous solution is the part with the const.

Demonstration: $y_p = \frac{1}{2}$ $y_h = C e^{-\frac{1}{2} t^2}$

Remember: $L(y) = t$ or $L(y_h + y_p) = t$

$$L(y_h) = 0 \quad \& \quad L(y_p) = t$$

The linear operator L is $\left(\frac{d}{dt} + 2t\right)$, so

$$L(y_p) = \frac{d}{dt}(y_p) + 2t y_p = \frac{d}{dt}\left(\frac{1}{2}\right) + 2t\left(\frac{1}{2}\right) = 0 + t = t \quad \checkmark\checkmark$$

$$L(y_h) = \frac{d}{dt}(e^{-t^2}) + 2t e^{-t^2} = -2t e^{-t^2} + 2t e^{-t^2} = 0 \quad \checkmark\checkmark$$

The homogeneous and particular solutions are built-in! (!)

Ex// #10 §2.2 $\cos(t) \frac{dy}{dt} + y \sin(t) = 1$

Put into right form: $\frac{dy}{dt} + \tan(t) y = \sec(t)$

Integrating factor: $\mu = e^{\int \tan(t) dt} = e^{\ln(\sec(t))} = \sec(t)$

Then $\frac{d}{dt}(\sec(t) y) = \sec^2(t)$

$$\sec(t) y = \int \sec^2(t) dt + C$$

$$y = \frac{1}{\sec(t)} [\tan(t)] + \frac{C}{\sec(t)}$$

$$y = \sin(t) + C \cos(t) \quad \checkmark\checkmark$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ y_p & + & y_h \end{array}$$

Ex// IVP's are also easy to deal with!

#18 §2.2

$$\frac{dy}{dt} - \frac{3}{t} y = t^3, \quad y(1) = 4$$

$$\mu = e^{\int -\frac{3}{t} dt} = e^{-3 \ln(t)} = \frac{1}{t^3}$$

$$\therefore \frac{d}{dt} \left(\frac{1}{t^3} y \right) = 1$$

$$\frac{1}{t^3} y = \int 1 dt + C$$

$$y = t^4 + C t^3, \quad \text{and now apply IVP!}$$

$$4 = (1)^4 + C(1)^3$$

$$C = 3, \quad \text{and plug back in!}$$

$$y = t^4 + 3t^3 \quad \checkmark \checkmark$$

Ex// #20 §2.2

$$(1+e^t) \frac{dy}{dt} + e^t y = 0$$

$$y(0) = 1$$

This can be solved easily w/ some ingenuity, if you're skilled at integrating.

$$\text{Divide: } \frac{dy}{dt} + \frac{e^t}{1+e^t} y = 0, \quad \mu = e^{\int \frac{e^t}{1+e^t} dt} = e^{\ln(1+e^t)} = 1+e^t \quad (!)$$

$$\therefore \frac{d}{dt} \left((1+e^t) y \right) = 0$$

$$(1+e^t) y = C$$

$$y = \frac{C}{1+e^t}$$

IVP time! $1 = \frac{C}{1+e^0} \Rightarrow C = 2$

$$\therefore y(t) = \frac{2}{1+e^t} \quad \checkmark \checkmark$$